League win prediction: model comparison

Group 27 | November 10, 2026 | Brief for TA/team discussion

Finding: the tested random forest performs similarly to logistic regression. This run gives no clear model-family winner; the final test set remains reserved.

Setup and design choices

Same 6,915 training and 1,482 validation matches, with 1,482 matches reserved for final testing. Core inputs are seven blue-minus-red economy/combat differences. Expanded inputs add four objectives/vision differences.

Forest: 500 trees, maximum depth 8, minimum 10 samples per leaf, square-root feature subsampling, bootstrap samples, Gini splits, seed 42. Settings were fixed before viewing forest scores. No parameter search, cross-validation or probability calibration was run. Logistic settings and match splits are unchanged.

Validation results

Model	Features	Accuracy	ROC-AUC	Brier	Log loss
Logistic regression	Core (7)	71.93%	0.7985	0.18362	0.54331
Random forest	Core (7)	71.79%	0.7965	0.18437	0.54571
Logistic regression	Expanded (11)	72.13%	0.8004	0.18285	0.53990
Random forest	Expanded (11)	71.86%	0.8004	0.18278	0.54012

Lower Brier/log loss is better. Majority accuracy is 50.07%; the constant training-rate baseline has log loss 0.69315. The earlier gold-only logistic model reached 71.05%; logistic core + objectives reached 72.27% (highest observed accuracy, without establishing a final winner).

What the differences mean

On matched features the forest makes 2 fewer correct decisions for core and 4 fewer for expanded inputs. Forest-minus-logistic accuracy intervals include zero: core **-0.13 points [-1.08, +0.74]**; expanded **-0.27 points [-1.28, +0.74]**. Expanded-model probability scores are almost identical across families.

Within the forest, expanded features improve log loss by **0.00560** but accuracy by only one correct match. The log-loss difference has a conditional 95% interval of [-0.01072, -0.00075]. This adds objectives and vision together; it does not identify which group helps.

Intervals use 2,000 paired validation-row bootstrap resamples. They hold fitted models fixed, do not adjust for multiple comparisons, and omit training/split variability and possible player dependence. One fixed forest configuration cannot rule out benefits from tuning or other nonlinear models.

Questions for the TA/team

1. Is the linear/nonlinear comparison, feature ablation and probability evaluation an appropriate progress-report scope?
2. Should we next prioritize training-only cross-validation and modest tuning, or richer data/features?
3. Which provenance/vision-quality limitations and data-use evidence should be resolved before final evaluation?

The current scores alone do not demonstrate that more data is required. The original CSV omits dates, patches, region and player IDs. No new data was collected, no message sent, and no assignment submitted. Full settings, predictions, uncertainty and reproducible code are included in the comparison package.